

4. SYSTEM ANALYSIS AND CLASSIFICATION

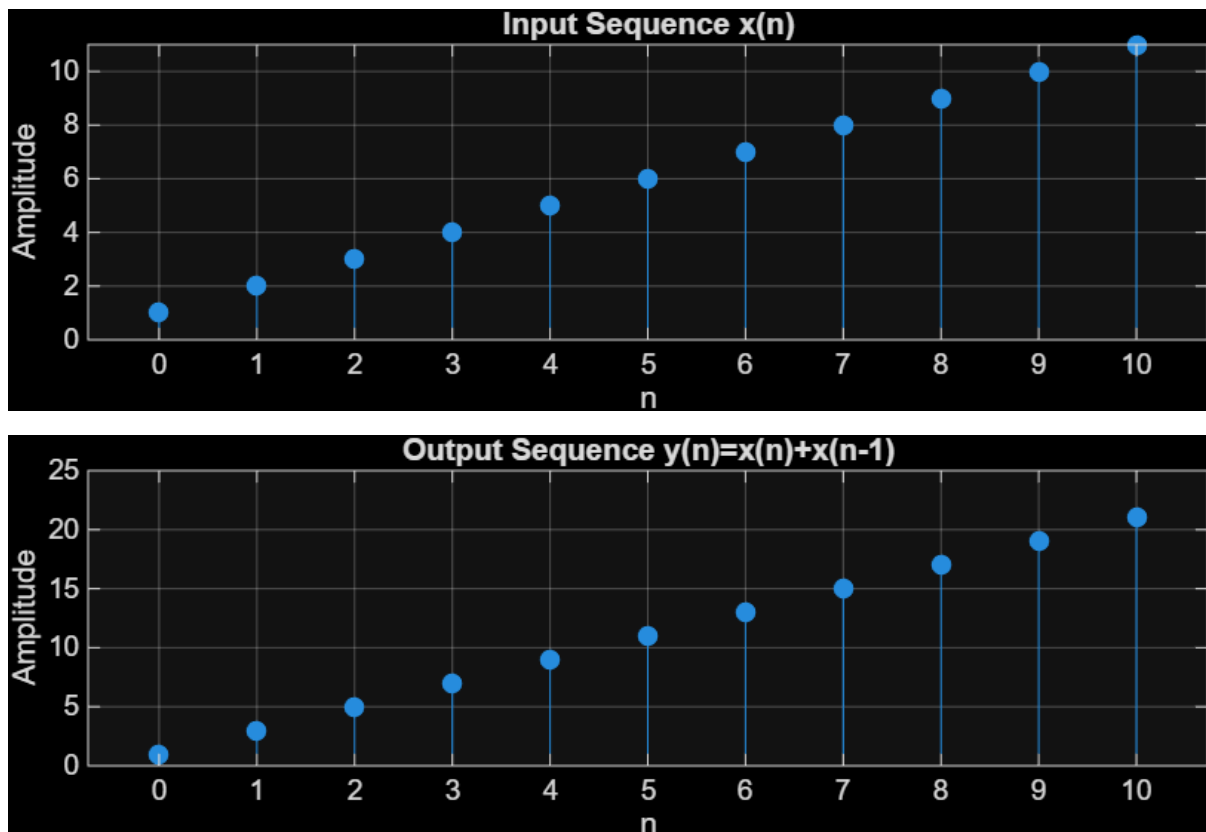
Experiment 4.1:

Discrete -Time System

Program:

```
clc;
clear;
close all;
% Input sequence
n = 0:10;
x = [1 2 3 4 5 6 7 8 9 10 11];
% Discrete-Time System
y = filter([1 1],1,x);
% Display output
disp('Output Sequence:');
disp(y);
% Plotting
figure;
subplot(2,1,1);
stem(n,x,'filled');
grid on;
title('Input Sequence x(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
stem(n,y,'filled');
grid on;
title('Output Sequence y(n)=x(n)+x(n-1)');
xlabel('n');
ylabel('Amplitude');
```

Output Waveform:



Result:

The discrete-time system $y(n)=x(n)+x(n-1)$ was implemented using MATLAB. The output sequence was obtained by combining the present and previous input samples, and the input and output signals were displayed using stem plots.

Experiment 4.2 :

Causal/Non-Causal System

Program:

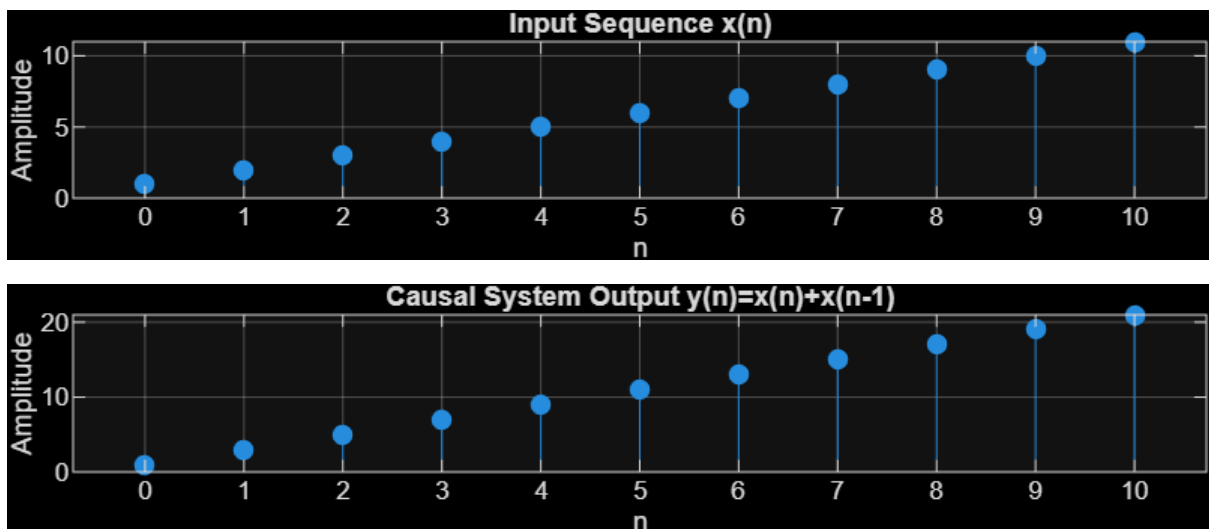
```
clc;
clear;
close all;
% Input sequence
n = 0:10;
x = [1 2 3 4 5 6 7 8 9 10 11];
% Causal System:  $y(n)=x(n)+x(n-1)$ 
y_causal = filter([1 1],1,x);
% Non-Causal System:  $y(n)=x(n)+x(n+1)$ 
```

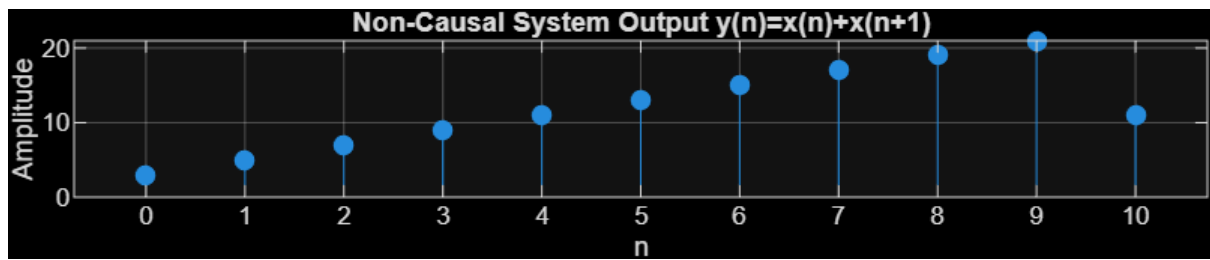
```

y_noncausal = x + [x(2:end) 0];
% Plotting
figure;
subplot(3,1,1);
stem(n,x,'filled');
grid on;
title('Input Sequence x(n)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,2);
stem(n,y_causal,'filled');
grid on;
title('Causal System Output y(n)=x(n)+x(n-1)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,3);
stem(n,y_noncausal,'filled');
grid on;
title('Non-Causal System Output y(n)=x(n)+x(n+1)');
xlabel('n');
ylabel('Amplitude');

```

Output Waveform:





Result:

The causal and non-causal discrete-time systems were implemented successfully using MATLAB. The causal system used present and past input samples, whereas the non-causal system used present and future input samples. The corresponding outputs were obtained and displayed using stem plots.